

FOOD PRICE STABILITY

PREDICTIVE ANALYTICS

Forecasting Key Commodity Prices Across Kenya
Using Machine Learning & Time Series Analysis



Dave
Ndung'u
—
Machine
Learning
&
Modelling



Trevor
Obonyo -
Non-
technical
Presenta
tion



Mading
Garang —
Machine
learning
&
Modelling



Leila
Abdikari
m —
Explanat
ory Data
Analysis



Clive
Kinyanjui
-
Business
Intelligen
ce Tools

The Problem: Food Price Volatility in Kenya

13.8%

Food Inflation
(June 2022)

67%

Maize Flour
Price Increase

Affected Items

- Wheat
- Cooking Oil
- Maize Flour
- Milk
- Potatoes
- Onions

These unpredictable fluctuations create a destabilizing cycle — oscillating between wasteful surpluses and acute shortages — affecting every Kenyan household.

Project Objectives

PRIMARY: Develop a predictive analytics model that forecasts prices of key food commodities — maize, milk, and beans — across Kenya.



Analyze historical food price trends across 47 counties



Study the effect of rainfall patterns on food prices



Measure the impact of fuel prices on transportation & food costs



Build & compare multiple machine learning models



Deploy an interactive Streamlit dashboard for live forecasting

Beneficiaries?

Consumers

Use price predictions to plan household budgets and prepare for food cost changes before they happen.

Government Agencies

Ministry of Agriculture & NCPB use insights for food security monitoring and policy decisions.

Farmers & Saccos

Decide when to plant, harvest, store, or sell produce based on predicted future market prices.

Insurers

Agricultural risk assessment and pricing strategies backed by predictive data.

DATA OVERVIEW

Three datasets merged into one powerful analytical foundation

Our Data Sources at a Glance

Food Prices

19,005 rows

Period: 2006 – 2026

- 27 Kenyan counties
- 51 commodities
- Retail & wholesale prices

Source: WFP

Rainfall

75,656 rows

Period: 1981 – 2026

- 10-day dekadal readings
- All 47 counties
- CHIRPS satellite data

Source: FEWS NET

Fuel Prices

197 rows

Period: 2010 – 2026

- Petrol, Diesel, Kerosene
- Nairobi pump prices
- Monthly records

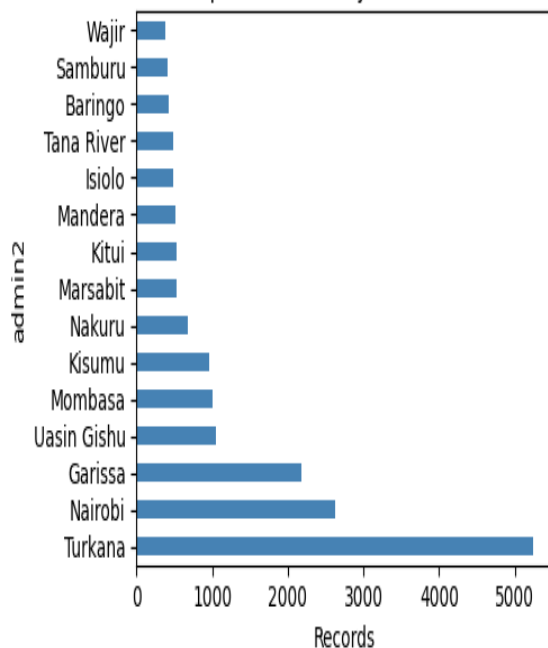
Source: EPRA

→ Merged into one unified dataset of 3,281 records (2010–2026)

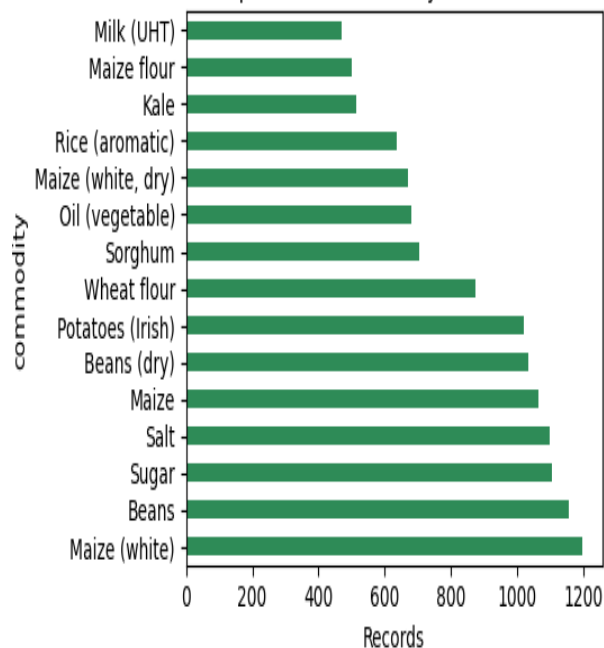
Data Coverage: Counties, Commodities & Price Types

WFP Food Prices – Overview

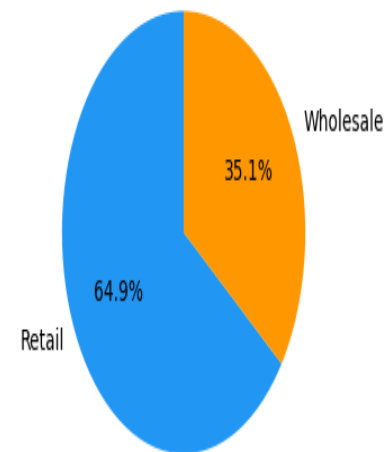
Top 15 Counties by Record Count



Top 15 Commodities by Record Count



Wholesale vs Retail Split

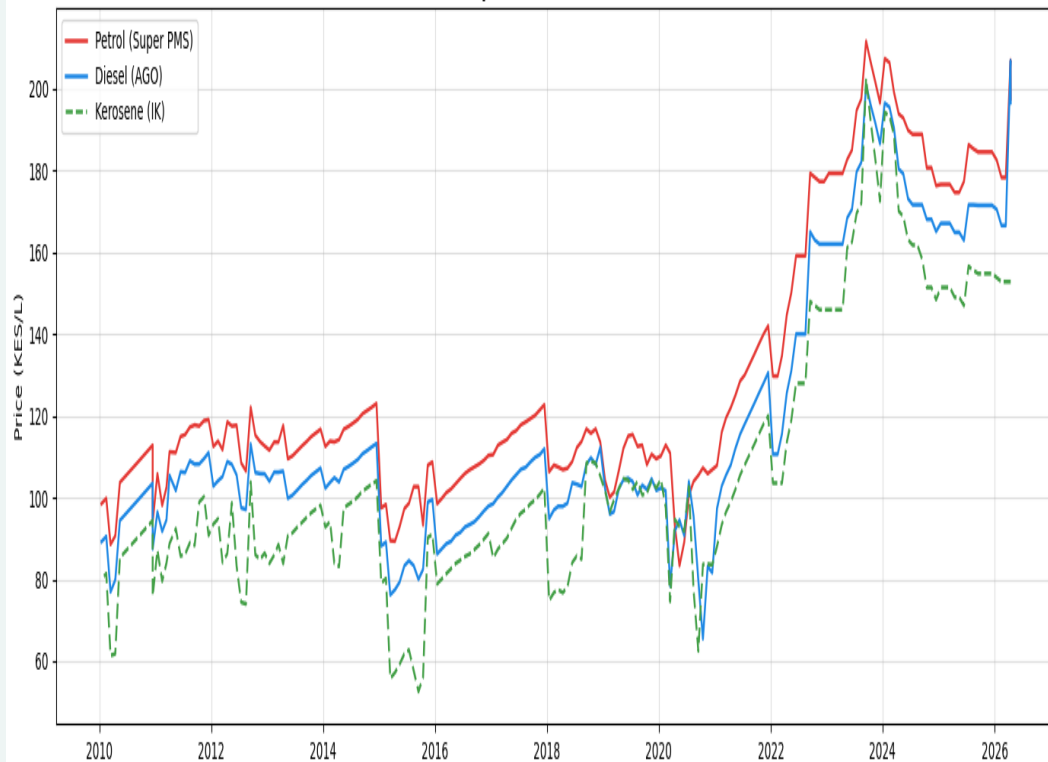


EXPLORATORY DATA ANALYSIS

What the data reveals before we build any models

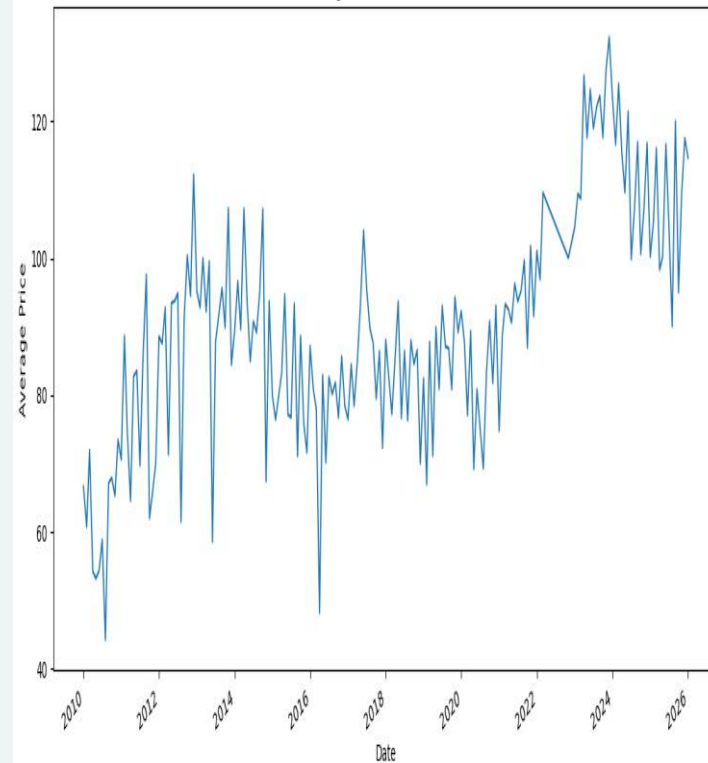
Fuel Prices Doubled; Food Prices Followed Suit

EPRA Pump Prices – Nairobi (KES/Litre)



EPRA Nairobi pump prices (2010–2026) →

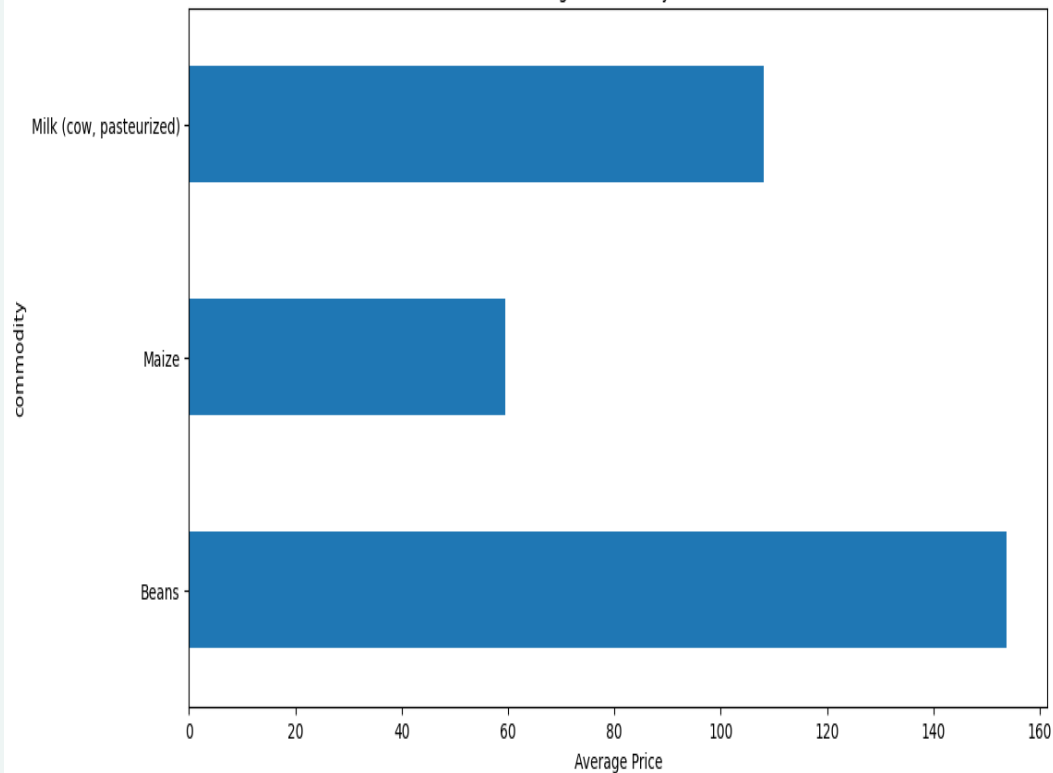
Average Food Prices Over Time



Average monthly food price →

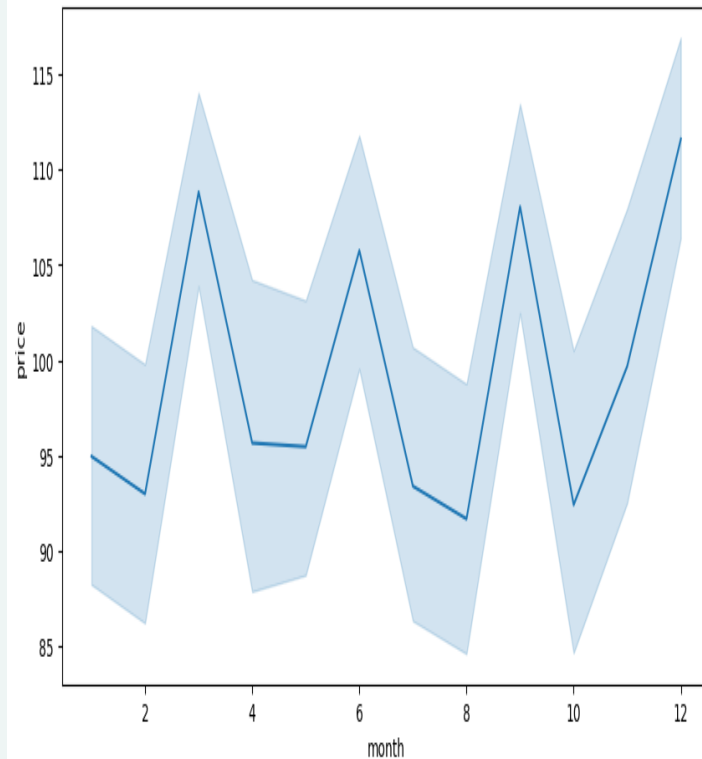
Commodity Prices & Seasonal Patterns

Average Commodity Prices



Average retail price by commodity →

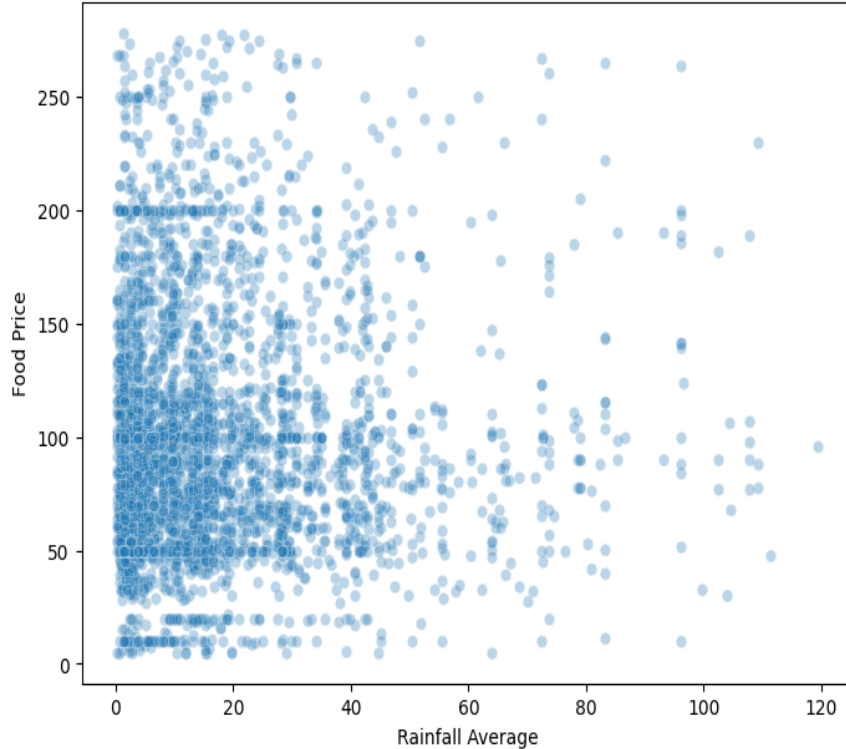
Seasonal Food Price Trends



Monthly seasonal price cycle →

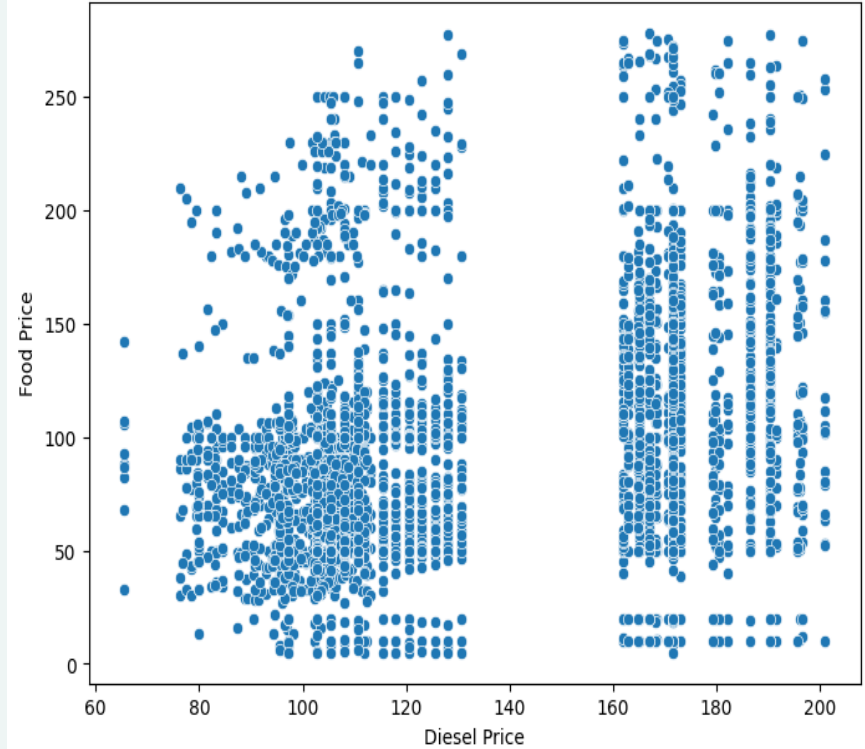
Drivers of Food Prices: Rainfall & Fuel

Rainfall vs Food Prices



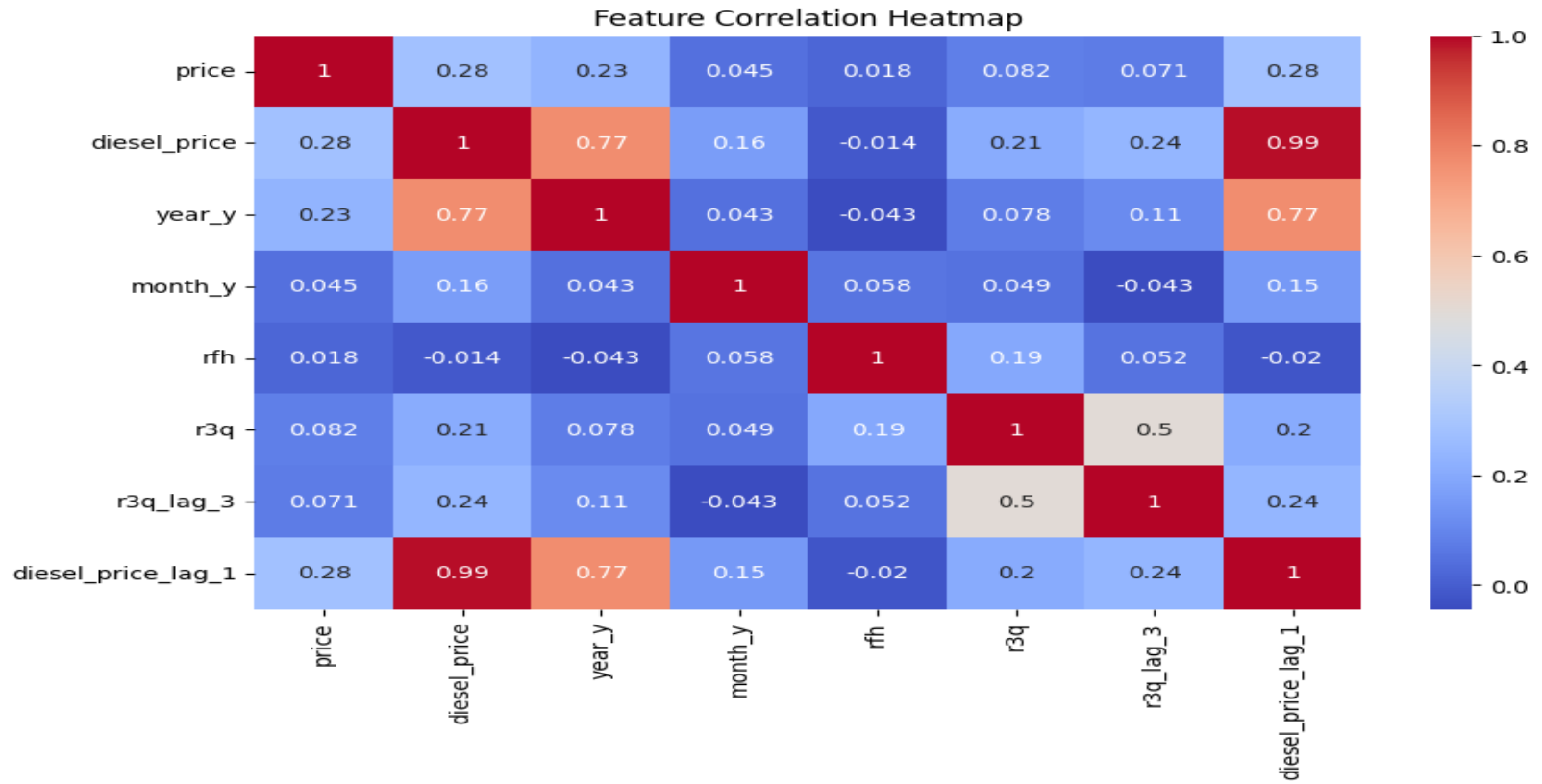
Rainfall (mm) vs food price →

Fuel Prices vs Food Prices



Diesel price (KES/L) vs food price →

Feature Correlation: Fuel Prices Dominate the Signal



Strong positive correlation between diesel price and food prices confirms fuel as a primary driver of food inflation.

Key Insights from the Data Analysis

Prices Rising

Food prices have risen consistently over 15 years, accelerating sharply post-2020.

Seasonality Matters

Clear monthly patterns emerge — harvest seasons push prices down, lean seasons push them up.

Fuel Is Key Driver

Strong correlation between diesel prices and food prices confirms fuel as the primary inflation lever.

Geographic Variation

Prices differ sharply across counties — remote regions like Turkana face consistently higher costs.

Rainfall Has Delayed Impact

Drought or excess rain today takes ~3 months to affect food prices — captured via time-lagged features.

Maize Is Central

Maize shows highest price volatility — making it the priority forecasting target.

PREDICTIVE MODELLING

Building & comparing multiple machine learning approaches

Our Modelling Strategy

TRACK 1: ML Regression

Predict prices using county, commodity, rainfall & fuel as inputs

Model 1: Linear Regression (Baseline)

Model 2: Random Forest (100 trees)

Model 3: XGBoost (Gradient Boosting)

GridSearchCV & RandomizedSearchCV tuning

80/20 train-test split

TRACK 2: Time Series

Forecast maize prices using historical sequences and calendar patterns

Seasonal Decomposition (Trend + Season)

Stationarity tests: ADF & KPSS

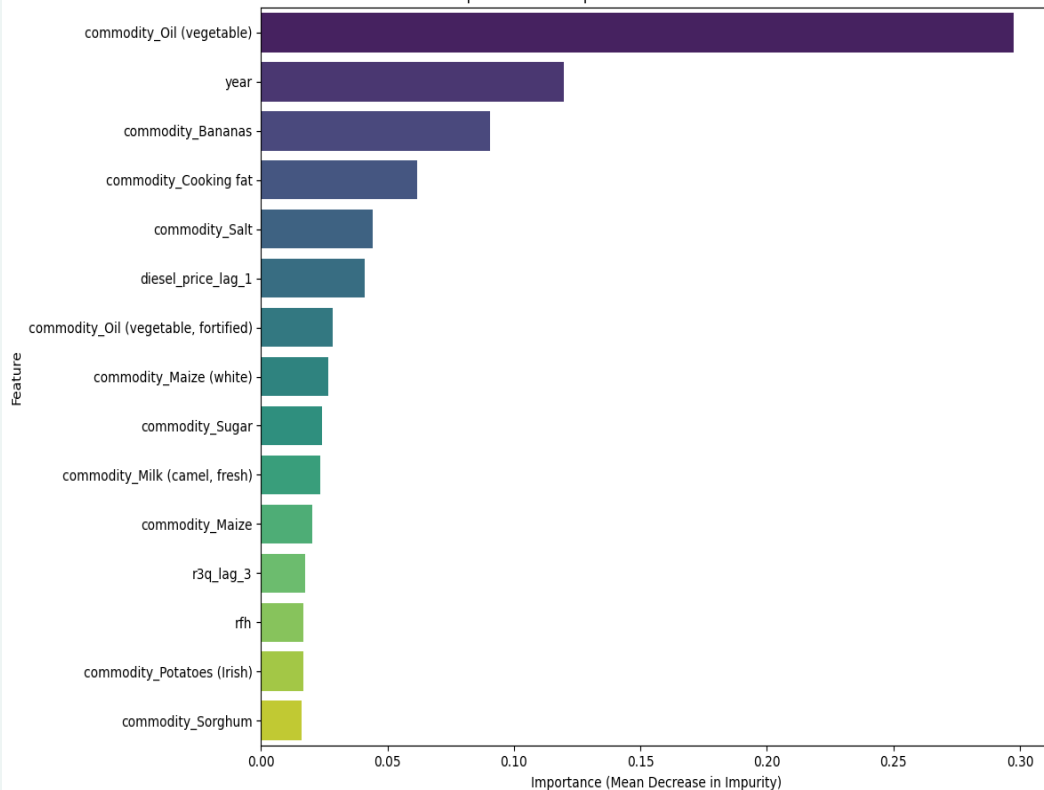
ARIMA: best model ARIMA(0,1,1)

SARIMA: seasonal variant (0,1,1)(0,0,0,12)

12-month ahead forecasting

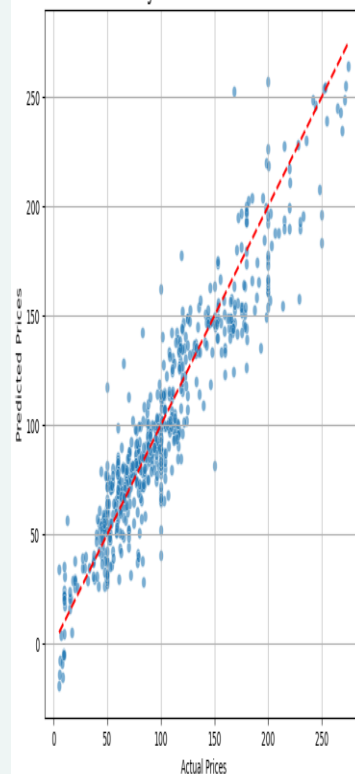
What Drives Prices & How Accurate Are the Predictions?

Top 15 Feature Importances from Random Forest Model

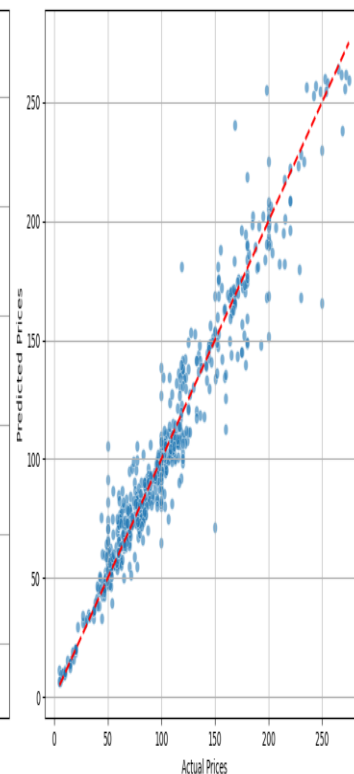


Top 15 features by importance (Random Forest) →

Linear Regression: Actual vs. Predicted Prices



Random Forest: Actual vs. Predicted Prices



Actual vs predicted prices (XGBoost) →

Model Performance: All Models Compared

Lower MAE / RMSE = More Accurate

Linear Regression (Baseline)

MAE: 15.07 KES

RMSE: 20.16 KES

R^2 : 0.86

Random Forest (Untuned)

MAE: 9.03 KES

RMSE: 14.96 KES

R^2 : 0.92

Random Forest (Tuned)

MAE: 10.75 KES

RMSE: 16.12 KES

R^2 : 0.91

XGBoost (Untuned)

MAE: 9.12 KES

RMSE: 14.56 KES

R^2 : 0.93

XGBoost (Tuned) ★

MAE: 8.65 KES

RMSE: 13.98 KES

R^2 : 0.93

ARIMA(0,1,1) Maize Only

MAE: 3.31 KES

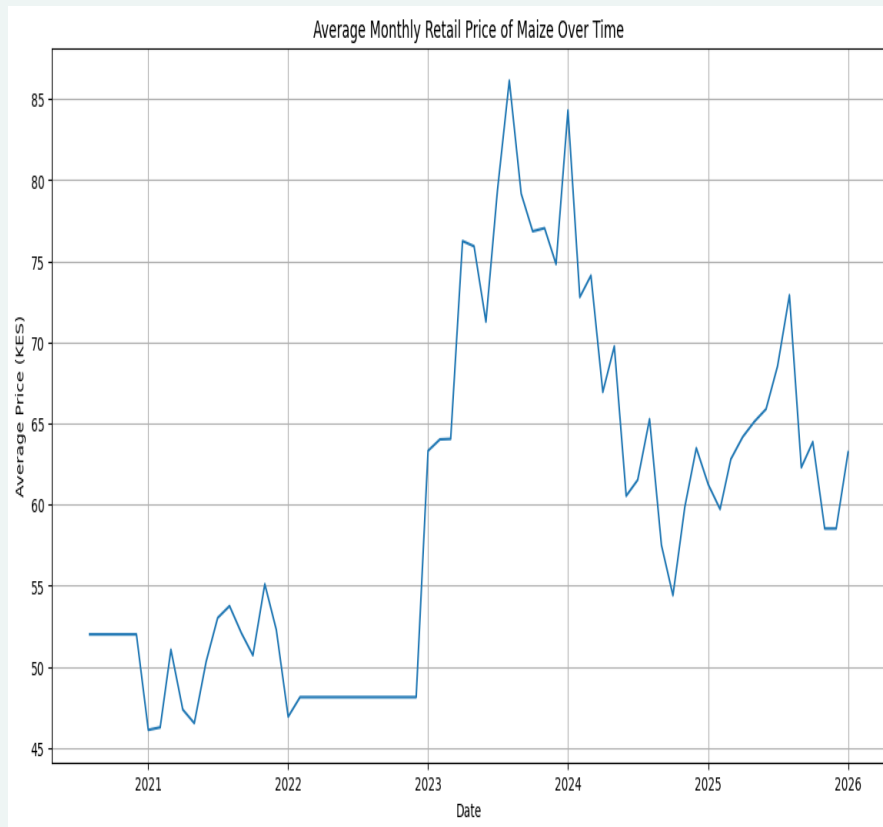
RMSE: 4.81 KES

R^2 : 5.28%*

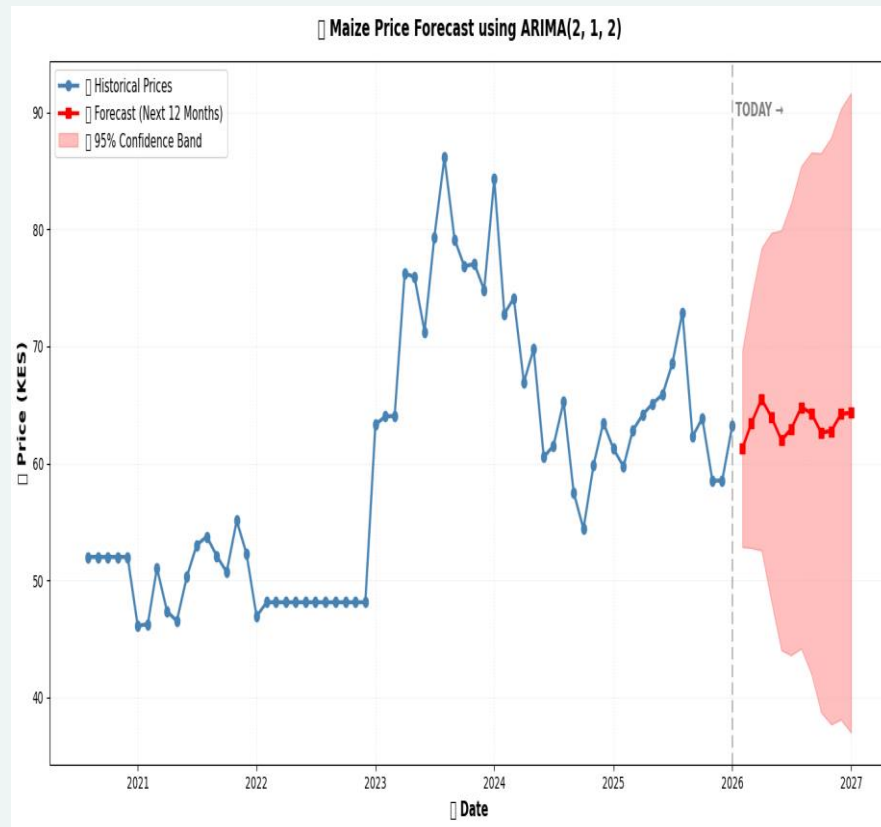
TIME SERIES FORECASTING

Predicting future maize prices month by month

Maize Price History & 12-Month ARIMA Forecast



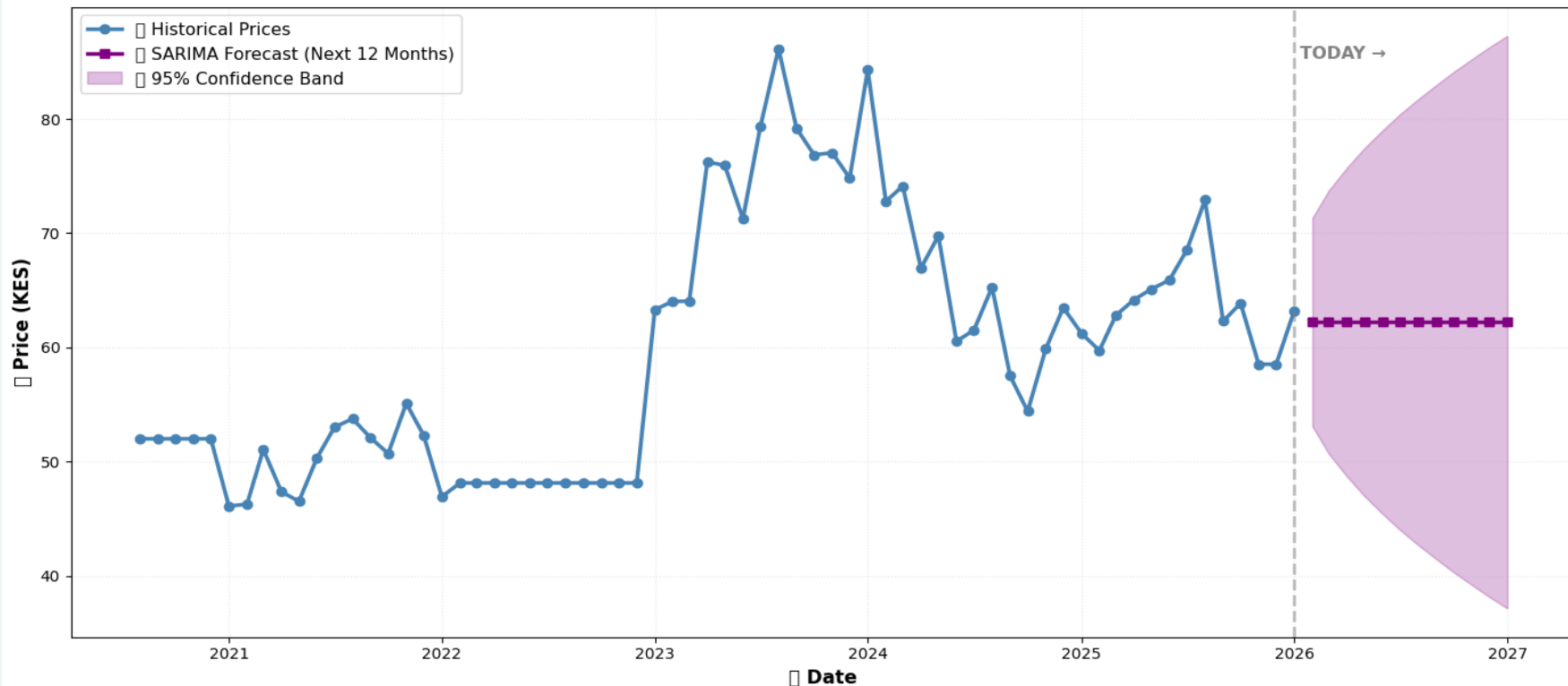
Historical monthly retail price →



ARIMA(0,1,1) 12-month forecast →

SARIMA Forecast: Capturing Seasonality in Maize Prices

Maize Price Forecast using SARIMA(0, 1, 1)(0, 0, 0, 12)



SARIMA(0,1,1)(0,0,0,12) — best seasonal model. Purple forecast at ~KES 62 with a widening confidence band over time.

Understanding ARIMA(0,1,1) in Plain Language

0 Autoregressive

We do NOT use past prices directly. Maize prices don't strongly depend on yesterday's value.

1 Differencing

We forecast price CHANGES rather than absolute prices. This removes the upward trend automatically.

1 Moving Average

We use last month's forecast error to correct this month's prediction — learning from mistakes.

3.31 KES

MAE

Average error

4.81 KES

RMSE

Weighted error

5.28%

MAPE

% error — excellent!

392.68

AIC

Lowest of 18 models

KEY FINDINGS & RECOMMENDATIONS

Turning data insights into action

What We Found: Key Project Findings

01

Food prices have risen significantly, with maize flour recording the sharpest increase at 67% in 2022.

02

Diesel price is the strongest predictor of food price — fuel cost rises directly translate to food inflation.

03

Rainfall impacts food prices with a 3-month delay — drought effects are slow but measurable.

04

XGBoost outperforms all regression models, with MAE of 8.65 KES and R^2 of 0.93.

05

ARIMA(0,1,1) delivers excellent time series forecasting for maize with 5.28% average error.

06

Clear seasonal patterns exist: prices drop after harvest months and spike in lean months.

Recommendations & Next Steps

Short-Term (0–3 months)

Update models monthly as new price data arrives • Deploy the Streamlit dashboard for public access • Expand commodity coverage

Medium-Term (3–12 months)

Add import/export volume data • Integrate climate-based early warning triggers • Pilot with farming cooperatives in Uasin Gishu

Long-Term (1+ years)

Partner with Ministry of Agriculture for official use • Connect to mobile money platforms for farmer alerts • Extend to East African regional markets

Thank You

Data-driven decisions for a food-secure Kenya.